

Challenge 3: AI PV Assistant

“Which rules really apply, and how can I plan my energy project better?”

Background

Anyone who wants to plan a PV system, battery storage system, wallbox, or heat pump today encounters contradictory information online: outdated press articles, one-sided provider websites, forum posts, political announcements, and actually applicable rules often appear side by side. At the same time, the EEG, EnWG, Metering Point Operation Act, technical connection requirements, grid operator specifications, funding programs, and municipal rules are difficult even for specialist companies to keep track of over time.

This creates uncertainty rather than implementation. Users often do not know which rules actually apply, whether registration is required, what funding or remuneration may be available, which technical decisions must be made, or which questions they should ask an installation company.

The goal of the challenge is an independent AI PV assistant that brings together current regulations, building data, and location data and uses them to provide understandable, source-based guidance — not as a sales tool, but as support for better decisions and qualified participation in shaping one’s own energy project.

Task

Develop an AI-supported prototype as a website or web app that connects regulatory information with building, location, and consumption data and derives an understandable initial assessment for decentralized energy projects.

The assistant should answer typical questions, for example:

- May I install a PV system on my roof?
- Which rules apply to storage systems, wallboxes, or heat pumps?
- Do I need to register anything with the grid operator?
- Which technical or regulatory requirements are relevant?
- Which funding, remuneration, or direct marketing logic might fit?
- Which decisions do I need to make, and which questions should I ask an installation company?

The prototype should transparently show current sources and legal statuses, name uncertainties, and help users better assess offers, formulate their own requirements more clearly, and prepare next steps. An optional extension is the automatic creation of a structured request (tender) for installation companies.

Available data basis

- Legal texts and regulations, e.g., EEG, EnWG, MsbG, state building regulations
- technical connection requirements and grid operator information
- funding and remuneration information
- Google Solar API, local solar cadastres, and building data

- user inputs on roof area, orientation, shading, consumption, wallbox, heat pump, storage system, and other consumers

Additional assumptions, creative visualization ideas, and extensions are expressly encouraged.

Expected prototype

The prototype should provide at least the following:

1. Evaluate current regulations

The assistant should process regulatory information from reliable sources and make it usable.

This includes, for example:

- legal texts and legislative amendments
- EEG and relevant energy-law regulations
- technical connection requirements, where available
- VDE-related requirements or standards references, where usable
- funding programs
- grid operator information
- municipal requirements
- official FAQ or information pages from public authorities

Important: the AI should not simply formulate freely, but should visibly work on the basis of a structured knowledge base.

2. Generate source-based answers

The assistant's answers should be traceable.

For each central statement, the prototype should, where possible, show:

- which source the statement comes from
- the legal or version status of the source
- whether the information is currently valid, announced, expired, or unclear
- what uncertainties exist
- where a more in-depth review is necessary

The goal is a knowledge base that is better than simply trusting a (possibly hallucinated) AI answer.

3. Take the project context into account

The assistant should not only provide general answers, but should also consider the specific location and the specific building.

Without narrowing down the specific project context, there are far too many variants to keep track of. (Example: a user asks about expanding an existing PV system. Because rules for controllability and feed-in remuneration have changed several times in the past, the assistant should first ask for the commissioning date and capacity, and then provide the specific answer.)

Possible inputs are:

- address or building location

- roof area
- roof orientation
- roof pitch
- building type
- residential or commercial use
- annual consumption (electricity, heat, electric car)
- existing or planned PV system, wallbox, heat pump
- planned storage system
- planned wallbox
- planned heat pump
- grid operator
- municipal solar cadastre information

On this basis, the assistant should explain which decisions need to be made and which rules and steps are likely to be relevant for this specific project.

4. Provide an initial technical assessment

In addition to regulatory guidance, the prototype should enable a simple preliminary technical check.

For example:

- rough PV suitability of the roof
- possible PV capacity based on roof area and orientation
- rough annual generation
- sensible combination with storage system, wallbox, or heat pump
- information on self-consumption and feed-in
- possible bottlenecks or open points for review
- initial prioritization of the next planning steps
- showing which decisions need to be made

The technical assessment does not have to be exact. It should serve as initial guidance for users and specialist companies and as a basis for deeper information.

5. Specific challenge requirements

In any case, the prototype should meet the following requirements:

Requirement 1:

For at least one concrete use case, the assistant must generate a source-based answer with reference to applicable regulations. The prototype may be limited to a clearly defined example context, e.g., a single-family home with a specific grid operator.

Requirement 2:

The assistant must incorporate user inputs (e.g., building data or location data) and derive a project-specific assessment from them.

Requirement 3:

The assistant must be able to distinguish between currently valid rules, announced changes, and outdated information.

6. AI explanation

The assistant should explain its recommendation in an understandable way. It may be assumed that users have an increased interest in understanding more complex relationships as well. Nevertheless, it must be possible to understand the explanations even with rudimentary, incomplete prior knowledge of energy law or technology (e.g., through tooltips, context-based additional explanations, etc.). The AI could also learn in which subject areas users want which level of detail.

Example:

“For this building, a PV system appears fundamentally sensible because the roof area is large enough and the orientation suggests good generation. For the planned 22 kW wallbox, prior approval from the grid operator must be obtained. The answer is based on the currently stored regulations and grid operator information. Two points are still open: the specific grid connection capacity and the desired metering concept.”

7. Planning portal or dashboard

Ideally, the prototype should function as a simple portal or object-centered dashboard.

Possible components:

- input form for address and project type
 - selection: PV, storage system, wallbox, heat pump
 - map or building visualization
 - rules checklist
 - source view
 - initial technical assessment
 - traffic-light or score system
 - next steps
 - export function
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8. Export

An exportable PDF one-pager should be generated for a project under consideration.

The one-pager should include:

- address or object ID
- planned measure, for example PV, storage system, wallbox, or heat pump

- brief initial technical assessment
 - relevant regulatory requirements
 - sources with legal or version status
 - open points for review
 - recommended next steps
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Data basis

The following are available or may be used:

- current legal texts
- legislative amendments
- EEG information
- EnWG information
- technical connection requirements
- relevant standards references
- funding information
- grid operator information
- Federal Network Agency
- Google Solar API
- local solar cadastres
- publicly accessible location and geodata
- user inputs
 - o building information
 - o roof area
 - o roof orientation
 - o roof pitch
 - o address
 - o building type
 - o existing or planned energy systems

The added value lies in bringing together regulatory knowledge, technical location data, and AI explainability in a simple planning assistant. This enables people to participate in shaping the energy transition on equal terms.

Not part of the challenge

The prototype does not have to:

- provide legally binding advice
- replace complete specialist planning
- carry out grid connection processes live
- automatically submit funding applications
- create binding profitability calculations
- fully reproduce paid standards texts
- generate real installation offers
- fully cover all grid operators in Germany